Aphid Screening Analysis

Setup

All genotypes in the Vicia faba PeaMUST collection

A. fabae datasets

There are:
- genotypes in the raw *A. fabae* dataset (no filters applied).
- genotypes in the filtered *A. fabae* dataset (after applying all filters).

A. pisum datasets

There are:
genotypes in the raw A. pisum dataset (no filters applied).
genotypes in the filtered A. pisum dataset (after applying all filters).

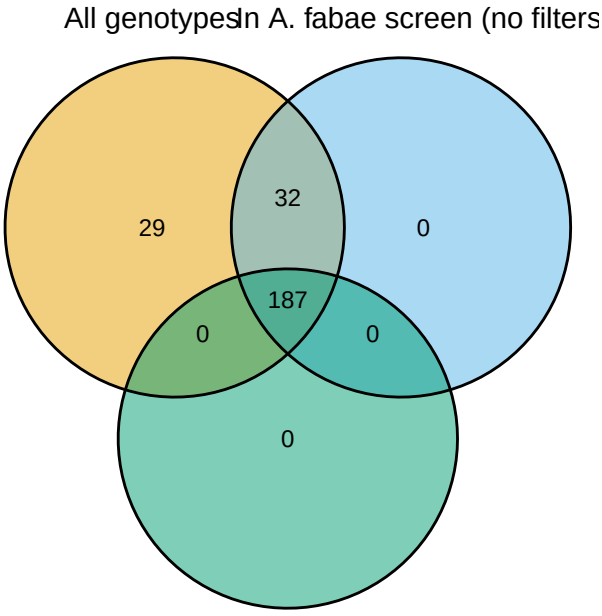
There are:
- genotypes in the raw A. pisum dataset (no filters applied).
- genotypes in the filtered A. pisum dataset (after applying all filters).

Venn diagrams

A. fabae

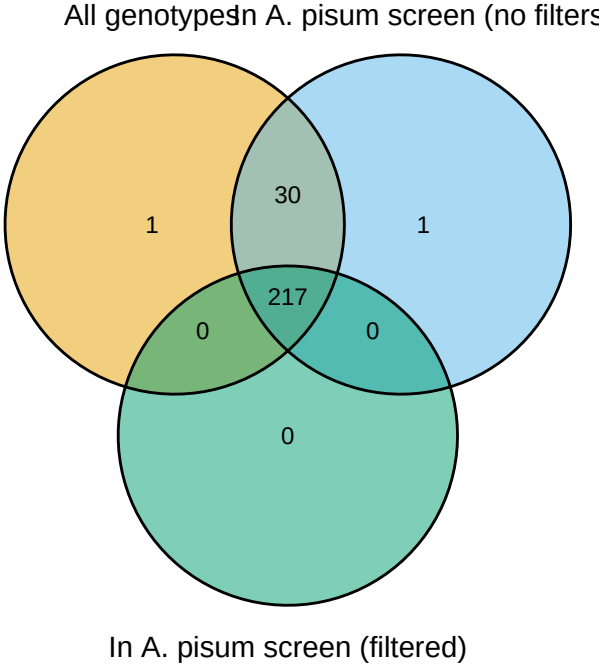
Create lists of genotypes from raw A. fabae data and after filtering.

Plot the Venn diagram to visualize the overlap between the three data sets.



A. pisum

Plot the Venn diagram to visualize the overlap between the three datasets.



Final list of missing genotypes

A. fabae

```
# List of missing genotypes
Afabae_missing_genotypes <-
  setdiff(
    genotypes_fabae_no_filters,
    genotypes_fabae_filtered) # A. fabae

afabae_missing_genotypes_df <- as.data.frame(Afabae_missing_genotypes)
afabae_missing_genotypes_df
```

```
   Afabae_missing_genotypes
1                 IVIP100127
2                 IVIP100225
3                 IVIP100236
4                 IVIP100252
5                 IVIP100279
6                 IVIP100319
7                 IVIP100328
8                 IVIP100375
9                 IVIP100387
10                IVIP100703
11                IVIP100878
12                IVIP100879
13                IVIP100922
14                IVIP100950
15                IVIP101216
16                IVIP101289
17                IVIP101371
18                IVIP101454
19                IVIP101496
20                IVIP101498
21                IVIP101536
22                IVIP101777
23                IVIP102063
24                IVIP102191
25                IVIP102220
26                IVIP102327
27                IVIP102335
```

```
28                IVIP102394
29                IVIP102405
30                IVIP102442
31                IVIP103046
32                IVIP103284
```

A. pisum

```
# List of missing genotypes
Apisum_missing_genotypes <-
  setdiff(
    genotypes_pisum_no_filters,
    genotypes_pisum_filtered) # A. pisum

apisum_missing_genotypes_df <- as.data.frame(Apisum_missing_genotypes)
apisum_missing_genotypes_df
```

```
   Apisum_missing_genotypes
1                 IVIP100120
2                 IVIP100199
3                 IVIP100227
4                 IVIP100320
5                 IVIP100325
6                 IVIP100437
7                 IVIP100633
8                 IVIP100796
9                 IVIP100891
10                IVIP101050
11                IVIP101179
12                IVIP101235
13                IVIP101309
14                IVIP101414
15                IVIP101451
16                IVIP101497
17                IVIP101508
18                IVIP101631
19                IVIP101748
20                IVIP101783
21                IVIP101791
22                IVIP101908
```

23	IVIP101913
24	IVIP101978
25	IVIP102080
26	IVIP102220
27	IVIP102305
28	IVIP102372
29	IVIP102732
30	IVIP103287
31	IVIP102346